

PROJECT REPORT OF CO-OP

Project at Industry (Module-2)

On

Integrating AI for Demand Forecasting in Online Marketplaces and ERP Systems

Submitted in partial fulfilment of the requirements for the award of degree

of

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In

COMPUTER SCIENCE AND ENGINEERING



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CERTIFICATE

This is to certify that the project report entitled "Integrating AI for Demand Forecasting in Online Marketplaces and ERP Systems" has been submitted for the Bachelor of Engineering (Computer Science and Engineering) at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, during the academic session January 2026 – May 2026.

This is a bonafide piece of work carried out by Vamika (2210992495) and Akriti Patial (2210991221), Department of Computer Science and Engineering, towards the partial fulfilment of the Integrated Project course, under the guidance of Prof. (Dr.) Ajay Kumar.

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CANDIDATE'S DECLARATION

We, Vamika (2210992495) and Akriti Patial (2210991221), B.E. 2022 batch, Department of Computer Science and Engineering, Chitkara University, Punjab, hereby declare that this project report is an original piece of work. The data and findings are authentic to the best of our knowledge. This report has not been submitted to any other institute for the award of any degree.

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Place: Rajpura

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CONTENTS

S.No.	Title	Page No.
	Acknowledgement	iii
	Abstract	iv
1	Introduction	1
2	Literature Review	2
3	Methodology	3
4	System Design and Implementation	4
5	Results and Analysis	5
6	Discussion	6
7	Conclusion and Future Scope	7
	References	8

ABSTRACT

Traditional ERP systems rely on historical data and fixed rules that fail to keep pace with volatile digital markets, resulting in inventory imbalances and missed revenues. This project proposes a Hybrid Integration Framework that combines internal ERP sales data with real-time external marketplace signals — competitor pricing and consumer trends — and applies interpretable Machine Learning models (Linear Regression and Random Forest) to improve demand forecasting. Results show a reduction in forecasting error (MAPE) from 26.4% to 7.8% — an 18.6% improvement — without requiring expensive deep-learning infrastructure. The framework is lightweight, scalable, and designed for small and medium enterprises (SMEs).

Keywords: *Demand Forecasting, ERP Systems, Machine Learning, Marketplace Data, Inventory Management, Random Forest, Linear Regression, Hybrid Integration Framework.*

1. Introduction

1.1 ERP Systems and Business Operations

Enterprise Resource Planning (ERP) systems integrate all core business functions — purchasing, manufacturing, sales, and HR — into a single platform. In a globalized economy, they are evaluated on their ability to maintain lean inventory while achieving near-perfect (99%) order fulfilment .

1.2 Demand Forecasting and Inventory Management

Forecast accuracy is directly tied to Economic Order Quantity (EOQ). Overestimating demand creates dead stock that ties up capital; underestimating leads to stockouts, lost sales, and brand erosion .

1.3 Limitations of Traditional ERP Forecasting

Most ERP systems use Univariate Time-Series methods such as Simple Moving Average (SMA) or Weighted Moving Average (WMA), which assume the future is a linear extension of the past . These are reactive — they lag behind real trends — and ignore external demand signals like competitor promotions or viral market trends.

1.4 AI and Machine Learning in Demand Forecasting

ML models shift forecasting from reactive to predictive. Through Multivariate Analysis, they simultaneously process internal sales history alongside external marketplace indicators (pricing, seasonality, consumer trends) to produce far more adaptive predictions .

1.5 Research Problem and Contribution

This project proposes a Hybrid Integration Framework that bridges the internal ERP SQL database with External Marketplace APIs, eliminating the data silo problem. The key contribution is a practical, interpretable, and scalable forecasting pipeline — designed for real-world ERP deployment — that improves forecast accuracy without deep-learning complexity .

1.6 Strategic Imperative and Scalability

Supply chain management is no longer a competitive advantage but a prerequisite for survival . The dual-layered architecture maintains ACID compliance internally (PostgreSQL) while elastically consuming external API data, making it deployable within standard enterprise infrastructure. The ultimate goal is to transform ERP from a passive "System of Record" into a proactive "System of Intelligence."

2. Literature Review

To provide context, the history of forecasting is examined across three phases.

2.1 Evolution of Time-Series Analysis

Classical methods — ARIMA (Box & Jenkins, 1970) and Exponential Smoothing (Gardner, 2006) — are foundational but confined to univariate data. Non-stationarity and seasonal volatility in retail expose their limitations .

2.2 Rise of Machine Learning in Supply Chain

Carbonneau et al. (2008) demonstrated that SVMs can outperform classical methods for non-linear

demand, particularly addressing the Bullwhip Effect. However, the absence of real-time external data remained a critical gap — one this project directly tackles.

2.3 Integration of External Indicators

Thiebaut et al. (2023) showed that marketplace sentiment analysis can improve short-term predictions by up to 12% . This project extends that finding by empirically validating competitor pricing as a second key external variable.

2.4 Identified Research Gaps

Most AI forecasting research uses opaque deep learning (LSTM) models that are difficult to integrate with standard ERP infrastructure . Separately, practical ETL documentation combining SQL (ERP) and JSON (Marketplace API) formats at an enterprise-framework level is extremely scarce — both gaps this project addresses.

2.5 Hybrid Frameworks and the Interpretability Crisis

Zhang et al. (2024) argue that hybrid, multi-layered data strategies resolve weaknesses of both pure statistical and pure ML models . In parallel, an "Interpretability Crisis" has emerged — industry practitioners often favour transparent sub-optimal models over black-box neural networks. This project uses Random Forest specifically because its built-in Feature Importance output gives supply chain managers a clear, auditable view of demand drivers .

3. Methodology

3.1 Data Sources and Collection

Two datasets were used in this study:

- Internal (ERP) Dataset: Four years of transactional records — SKU codes, unit prices, and weekly quantities sold — extracted from a PostgreSQL database.
- External Marketplace Dataset: API-pulled data from multiple e-commerce platforms, capturing market prices for matching SKUs and a Consumer Trend Score.

3.2 Data Preprocessing and Cleaning

Outlier detection using the Interquartile Range (IQR) removed data points skewed by temporary events. Price and quantity values were then normalized using min-max scaling so no single variable dominated model training .

3.3 Feature Engineering — Feature Fusion Layer

Three derived features were engineered to enrich model inputs:

- Price Elasticity Index – ratio of the organization's price to the average competitor price, capturing relative market positioning.
- Seasonality Lag – 12-month historical sales lag, encoding year-on-year demand patterns.

- Rolling Mean (7-day) – 7-day moving average to smooth short-term fluctuations while preserving trend signals.

3.4 Machine Learning Models

Three models were implemented and benchmarked:

- Baseline — Simple Moving Average (SMA): the standard conventional ERP method.
- Linear Regression (LR): captures the linear relationship between marketplace price signals and demand.
- Random Forest Regressor (RFR): an ensemble of decision trees chosen for non-linear modelling capability and its built-in Feature Importance ranking .

3.5 Evaluation Metrics

- Mean Absolute Error (MAE): average absolute deviation from actual demand values.
- Mean Absolute Percentage Error (MAPE): percentage error, enabling intuitive accuracy statements (e.g., "92% accurate") for business stakeholders .

4. System Design and Implementation

4.1 Proposed Architectural Framework

The system follows a 4-tier architecture:

- Data Layer: PostgreSQL ERP database + Web Scraper for Marketplace API data.
- Integration Layer: Python-based ETL pipeline that harmonizes SQL and JSON formats into a unified analytical dataset.
- Analytics Layer: Scikit-learn environment for model training and hyperparameter tuning.
- Presentation Layer: Streamlit dashboard for inventory managers — displays forecasts, demand trends, and feature importance rankings in real time.

4.2 Implementation — The Python Stack

The framework uses only open-source libraries to ensure SME accessibility:

- Pandas / NumPy: Data manipulation and matrix operations.
- Scikit-Learn: Random Forest and Linear Regression implementations with cross-validation.
- Matplotlib / Seaborn: Demand trend visualization and model comparison plots.
- Streamlit: Interactive management dashboard with no front-end engineering required.

4.3 Model Training and Validation

An 80/20 temporal split was used — 80% for training and the most recent 20% held out for testing — preserving time ordering to prevent data leakage. Random Forest hyperparameters (number of trees, max depth, minimum samples per leaf) were tuned via 5-fold cross-validated grid search.

5. Results and Analysis

5.1 Model Performance Comparison

Table 1 summarizes forecasting accuracy across all three models:

Model	MAE	MAPE (%)	Accuracy (%)
Simple Moving Average	142.3 units	26.4%	73.6%
Linear Regression	98.7 units	15.2%	84.8%
Random Forest	47.1 units	7.8%	92.2%

Table 1: Comparative model performance

5.2 Demand Trend Visualization

The Random Forest model correctly predicted the holiday demand surge in November approximately 14 days in advance. The Simple Moving Average only caught up after the peak — resulting in a theoretical stockout scenario with direct revenue impact. Graphical comparison confirms that Random Forest predictions closely track actual demand across both stable and volatile periods .

5.3 Feature Importance Analysis

The Random Forest's Feature Importance ranking identified the key demand drivers:

- Previous Month Sales — strongest predictor (internal ERP signal).
- Marketplace Price Index — second most important, confirming that competitor pricing is a critical signal that conventional ERP systems miss entirely.
- Consumer Trend Score and Seasonality Lag — strong secondary contributors, especially during promotional windows.
- Rolling Mean (7-day) — consistent noise-reduction benefit across all test periods.

6. Discussion

6.1 Interpretation of Findings

Price and seasonality interact non-linearly — a relationship Linear Regression cannot fully model. Random Forest handles these interaction effects effectively, producing the 18.6% MAPE improvement over the baseline. The model's decision-tree structure mirrors how a human planner reasons: evaluating multiple conditional factors simultaneously before committing to a forecast.

6.2 Practical Implications for Industry

For medium-sized businesses, an 18.6% forecast error reduction directly cuts costs from dead stock, emergency procurement, and stockout losses. Since the framework uses only open-source tools and standard APIs, these gains require no major capital expenditure.

6.3 Mitigation of the Bullwhip Effect

The Bullwhip Effect — where small demand errors at the consumer level amplify exponentially upstream — is significantly reduced by the synchronized, forward-looking demand signal this framework provides. Narrowing the gap between forecast and actual demand releases capital tied up in excess safety stock .

6.4 Democratization of Predictive Analytics

A central finding is that SMEs do not need deep-learning infrastructure for enterprise-grade forecasting. Prioritizing "feature quality" — choosing the right external data to include — outperforms increasing model complexity. This reframes the competitive advantage: from a computational arms race to a data strategy race .

7. Conclusion and Future Scope

7.1 Conclusion

This project validated that augmenting ERP systems with external marketplace signals and interpretable ML models produces significant forecasting improvement. Random Forest achieved 92.2% accuracy (MAPE: 7.8%) against the SMA baseline of 73.6% (MAPE: 26.4%) — an 18.6% gain. Beyond the numbers, the project demonstrates a viable path for transforming ERP from a passive "System of Record" into a proactive "System of Intelligence" — enabling just-in-time inventory strategies that respond to real market conditions rather than lagged internal transactions .

7.2 Future Scope

- Deep Learning Integration: LSTM networks for capturing long-horizon seasonal dependencies.
- Edge Computing: Deploying the pipeline on warehouse scanners for real-time in-situ calibration.
- NLP Integration: Quantifying customer review sentiments as additional model features.

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